Experts

1 Analytic Equations

The famous formula of Leibniz involving π is

$$\frac{\pi}{4} = 1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \frac{1}{9} - \cdots = \sum_{k \geq 0} \frac{(-1)^k}{2k+1}. \tag{1}$$

However, (1) converges far too slowly to be a practical method of approximating π.

Euler's integral formula for $n!$ is

$$n! = \int_0^\infty x^n e^{-x} dx.$$

The right side makes sense at $n = 1/2$ and can be evaluated explicitly:

$$\int_0^\infty \sqrt{x} e^{-x} dx = \frac{\sqrt{\pi}}{2}.$$

This suggests defining $(1/2)! = \frac{\sqrt{\pi}}{2}$.

For $a \in R$ and a function $f : \mathbb{R} \to \mathbb{R}$, the derivative of f at a is

$$\begin{aligned} f'(a) &= \lim_{h \to 0} \frac{f(a+h) - f(a)}{h} \\ &= \lim_{x \to a} \frac{f(x) - f(a)}{x - a} \end{aligned}$$

More formulas like this can be found in the book [**?**, Rud76].

2 Areas

If we are dealing with lattice polygons that are not simple, then we can use the following extension of Pick's Theorem.

Theorem 2.1. *For a lattice polygon P containing h holes with a boundary of $h+1$ simple closed curves, its area is given by*

$$A(P) = I(P) + \frac{B(P)}{2} + h - 1,$$

where $I(P)$ is the number of interior lattice points of P and $B(P)$ is the number of boundary lattice points of P.

We now review the definition of Euler characteristic for a convex polyhedron.

Defination 2.1. *The Euler characteristic of a convex polyhedron P with V vertices, E edges, and F faces is $\chi(P) = V - E + F$.*

Next, we recall Euler's formula.

Theorem 2.2. *(Euler's Formula). For any convex polygon P with Euler characteristic $\chi(P)$, we have that*

$$\chi(P) = 2. \tag{2}$$

We will now generalize Theorem 2.1 by incorporating the Euler characteristic, which will allow us to find the area $A(P)$ of any lattice polygon P that is the union of a finite number of simple lattice polygons. This more general area formula is given by

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix}$$